

VotaSJ: Trust, Counted Live

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Abstract. *VotaSJ is a blockchain-anchored participatory-budgeting platform for São José/SC. Citizens authenticate with gov.br, vote from their phone, and audit the tally and the budget execution in real time — without having to trust the City Hall to count its own vote. The platform replaces a BRL 300–500k physical process with a BRL 90–160k digital one, governed by a 3-of-5 multisig (City Hall, UFSC, TCE-SC) and a 14-day timelock on rule changes. This short paper makes the case to the institutional buyer (§) and to the citizen (§).*

For the Investor and the Municipality

Problem. Brazilian municipalities run participatory budgets that almost no one trusts. In São José/SC: turnout below 2 % of 180,000 voters (<3,600 people decide tens of millions of reais); operational cost of **BRL 300–500k/cycle** dominated by physical assemblies and ≈3-week manual tallying; and a tally run, counted, and announced by the City Hall itself, so every contested result ends in an unfalsifiable accusation of manipulation.

Solution. A platform on Polygon PoS where citizens authenticate with gov.br, vote from their phone, and audit results live. The blockchain holds only what the City Hall must not be allowed to rewrite — the eligibility commitment, the votes, the tally, and the budget-execution milestones. Personal data, proposal text, photos, and UX stay off-chain. Administration is a **3-of-5 multisig** (3 City Hall, 1 UFSC, 1 TCE-SC) and any contract change is gated by a **14-day timelock**, so the rules cannot move inside an active vote and the chain keeps running even if the City Hall stops cooperating.

Why now. gov.br is mature (≈150 M citizens, free for municipal integration); Polygon per-vote gas is under USD 0.005; OpenZeppelin + Hardhat make audited deployments affordable; TCE-SC and MP press for auditable processes — we arrive with the audit built in.

Market.

Segment	Scope	3y addressable
São José/SC (anchor)	≈180k voters	BRL 220k/yr
Greater Florianópolis	4 cities, ≈1.1M voters	BRL 0.8–1.2M/yr
Santa Catarina state	≈5.5M voters	BRL 4–6M/yr
BR cities >50k inhab.	≈340 cities, ≈85M voters	BRL 60M+/yr

TAM: >340 cities legally required to run participatory cycles. SAM: ≈80 with budgeted digital modernization. SOM (36 mo): single-digit municipalities anchored on São José.

B2G SaaS model. Annual subscription BRL 180–250k/city; per-cycle setup BRL 40k; implementation consulting BRL 60–120k; white-label for professional councils. Citizens

never pay. *Unit economics*: physical process **BRL 300–500k/cycle**; VotaSJ **BRL 90–160k/cycle, all-in**; net savings **BRL 170–300k/cycle**, before counting the political value of an unimpeachable result.

Traction. Production-ready platform — VoterRegistry, ParticipatoryBudget, and ExecutionTracker deployed under a UUPS proxy with a 3-of-5 multisig and 14-day timelock, exercised by a 40-test Hardhat suite, CI-enforced lint/coverage gates, $\approx 430k$ gas measured for a full citizen flow. Institutional partners: UFSC Blockchain Lab, São José Planning Secretariat, TCE-SC. Rollout: field pilot 2026-Q3, full São José cycle 2026-Q4, multi-city expansion 2027.

Ask. An institutional pilot agreement with São José/SC and a seed envelope of **BRL 400–600k** for the first external smart-contract audit, gov.br integration certification, and the field-pilot cycle. First paid contract closes 12 months after disbursement; payback is one cycle.

For the Citizen

Your vote, your eyes. Today, when you participate in São José’s participatory budget — if you participate — you go to a night-time assembly, raise your hand, and trust that the City Hall counts correctly and executes what won. VotaSJ changes one thing: **you can stop trusting and start checking.**

What changes for you.

- **Vote from your phone, on your own schedule.** No more assemblies you can’t attend because of work, kids, or distance.
- **See your vote land.** The moment you cast it, the public tally updates. Refresh the page and watch it.
- **Anyone audits the count.** Journalists, neighborhood associations, TCE-SC, MP — same data, same chain, same time.
- **Follow execution.** You’re notified when your winning proposal moves to contract, work start, milestones, and finish, with evidence you can check.
- **Your data stays private.** CPF, name, address: none of it goes on the blockchain. Only a one-way hash that proves you’re eligible, not who you are.

Three steps. (1) Sign in with gov.br — the federal ID you already use. (2) Pick a proposal: browse what neighbors submitted, read the budget, vote once. (3) Watch the result and the execution: real-time tally and per-milestone updates. No need to know what a “blockchain” is. No crypto, no fee, no extra app.

Forquilha, next cycle. Maria, a cashier whose shift ends at 22:30, opens VotaSJ on the bus home and votes for the renovated daycare. Months later she is notified — contract signed, work started, daycare reopens — each step with verifiable photographic evidence. 90 seconds of participation; the institution actually delivered.

Trust, but verify — and now you actually can.

Companion document and references. A 16-page technical white paper covers architecture, threat model, gas measurements, team, and roadmap. Sources: *Estatuto da Cidade* (https://www.planalto.gov.br/ccivil_03/leis/leis_2001/l10257.htm), *LGPD* (https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/lei/l13709.htm), *gov.br* (<https://acesso.gov.br/roteiro-tecnico/>), *Polygon PoS* (<https://docs.polygon.technology/>).